

Sunburn device: the

The Sunburn Flowchart, built from the real blocks of the program

The Sunburn device is a wearable built on a BBC micro:bit, a small computer with 25 LEDs where you are through a phone, and then flashes, beeps or taps your arm until you put built from 74 coloured blocks. Every one of them is on these sheets, with a note saying

Follow the arrows. Each red label is a box of the flowchart; the coloured picture under it is the actual block forever block runs this plan round and round, all day.

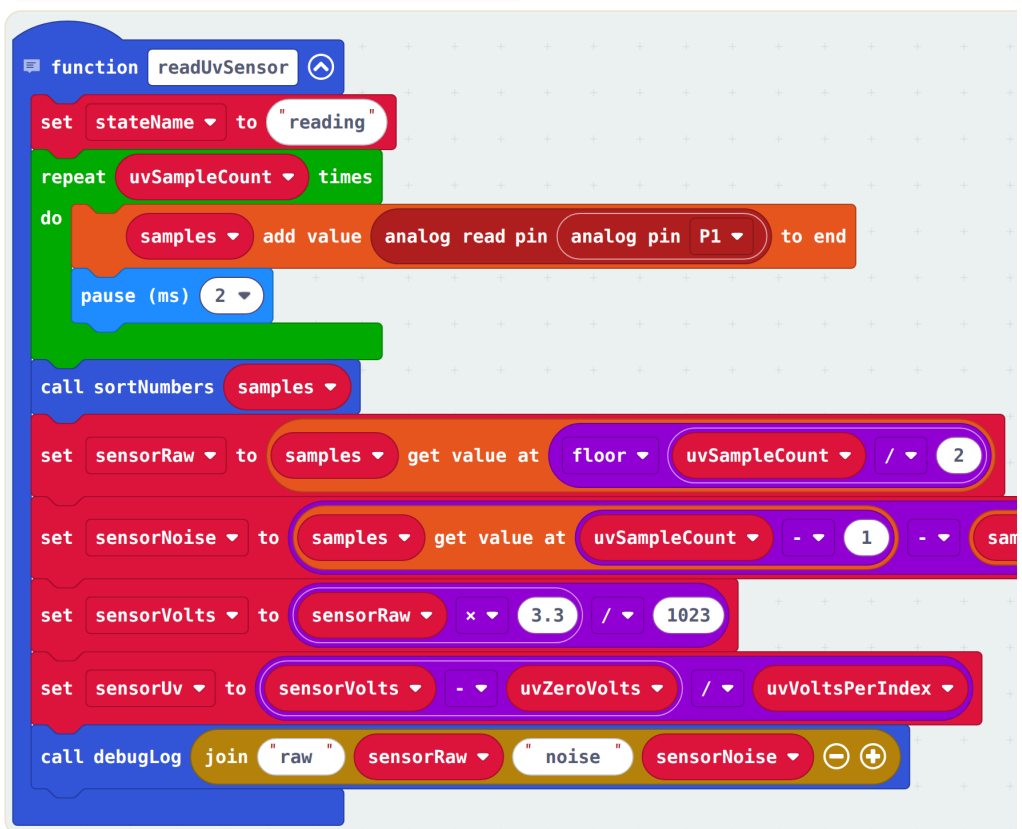
Device turns on



deviceTurnsOn

Switching on: shows a tick, says Device on on the screen, tells the app, and marks the device as on.

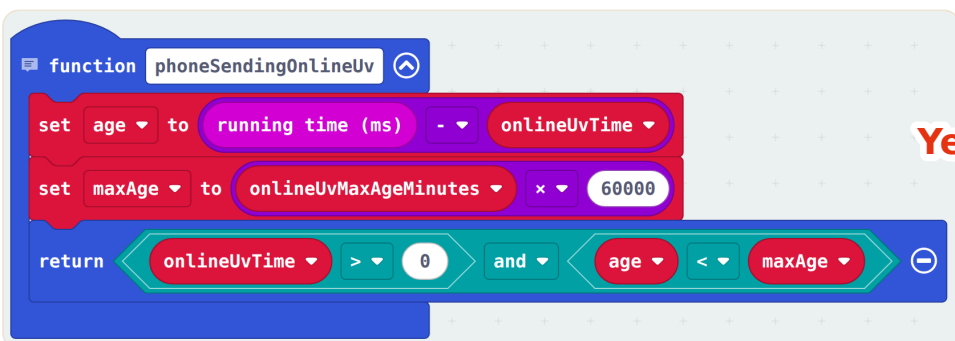
Read the UV sensor



readUvSensor

Reads the UV sensor plugged into pin P1. It takes 15 quick readings, sorts them, uses the middle one, so a single odd reading cannot set the value. Then it turns the sensor's voltage into a UV index number.

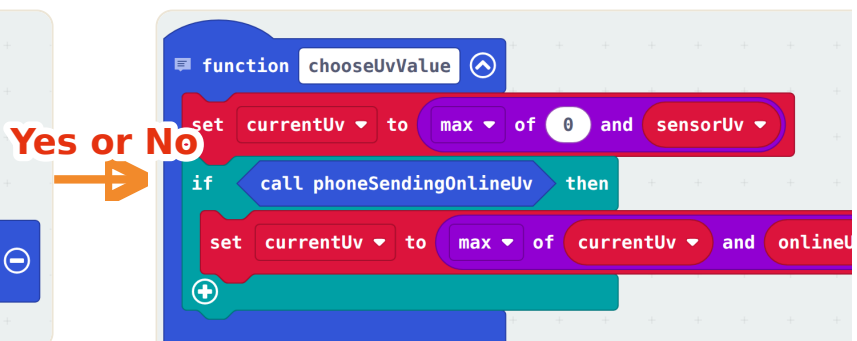
Is the phone sending the online UV index?



phoneSendingOnlineUv

Asks: has the phone app sent an online UV index in the last 20 minutes?

Pick the UV value: the highest of sensor and online



chooseUvValue

Picks the UV number the device will act on. It takes the highest of the sensor's reading and the online reading if it's available.

program on the wrist

am, with what each one does

hts and two buttons. It measures the sun's UV with a sensor, checks the online UV index for sunscreen on. Two hours later it reminds you to reapply. The program that does all this is what it does.

k of code that does that job, exactly as it looks in the MakeCode editor, and the note says what it does. The

try again

No: CHECK SENSOR, try again

Is the reading believable?

function sensorReadingInRange

set inRange to true

if sensorRaw ≥ 1020 then

set inRange to false

if sensorNoise > 200 then

set inRange to false

if sensorUv < -1.5 or sensorUv > 20 then

set inRange to false

if inRange then

set sensorErrorReported to false

return inRange

No

showCheckSensor

set stateName to error

if not sensorErrorReported then

call sendLine ev=error;msg=CHECK SENSOR

set sensorErrorReported to true

show icon

call showMessage CHECK SENSOR

call oledFooter join pin call formatNumber sensorVo

call beep 440 200

call beep 440 200

pause (ms) sensorRetrySeconds × 1000

sensorReadingInRange

Asks: is this reading believable? The answer is no if the sensor is missing or shorted, if the readings jump about (a loose wire), or if the number could not be real.

showCheckSensor

When the reading is not believable: sh and the words CHECK SENSOR, beeps the app there is a problem, and tries a seconds. A faulty sensor is never treat

How strong is the sun?



on:



The Sunburn Flowchart

gain

ain in 5 s

lts ▾

" V "

⊖ ⊕

ows a cross
twice, tells
gain in 5
ed as safe.

the last 30 minutes?

the higher of the sensor's reading and the phone's online value, or just the sensor reading when the phone is not sending.

0 to 2

0 to 2, low: happy face

```
function showHappyFace
  set stateName to "safe"
  show icon [happy face]
  call oledShowMessage "Low UV. No sunscreen needed"
```

showHappyFace
UV 0 to 2: shows a happy face. No sunscreen needed right now.

3 to 7, moderate or high: alert and slow beep until A

```
function moderateOrHighUv
  if protectionLevel = 0 then
    set stateName to "alert"
    call showMessage "Sunscreen time"
    call flashAndBeepUntilA false
    if ackPressed then
      call showMessage "1 tsp SPF 50+ on each arm, each leg"
      call sunscreenApplied 1
    else if deviceOn then
      call standbyUntilButton
    else
      call alreadyProtected
```

moderateOrHighUv
UV 3 to 7: flashes and beeps slowly until button A is pressed, then shows how much sunscreen to put on and starts the 2 hour timer. If already on, it just shows a tick and left. If nobody presses A within 10 minutes, the device assumes it is not being worn and goes to sleep.

sunscreen already on

A pressed

Sunscreen already on: just a tick

```
function alreadyProtected
  set stateName to "protected"
  show icon [tick]
  set timeLeft to reapplyDueTime - running time (ms)
  call oledShowMessage join "Sunscreen on. Reapply in " call formatTime timeLeft
```

alreadyProtected
Sunscreen is already on, so there is no alert: just a tick and the time until it needs reapplying.

A pressed on

```
function sunscreenOn
  show icon [sunscreen]
  set protectionLevel to 3
  change sunscreenOn to true
  call sendEvent "sunscreenOn"
  call start2HourReapply
```

sunscreenOn
A was pressed: remember the alert: remember the time given, countdown, tells the app.

then back to the top

then back to the top

```

set band to vhigh
+
return band

```

Turns a UV index into one of the four levels on the flowchart: low (0 to 2), moderate or high (3 to 7), very high (8 to 10) or extreme (11 and above).

11 and above

**8 to 10, very high: flash and fast
beep until A**

11 a
tap t

```
function veryHighUv
  if protectionLevel < 2 then
    set stateName to "alert"
    call showMessage "Very high UV"
    call flashAndBeepUntilA true
  if ackPressed then
    call showMessage "1 tsp SPF 50+ on each arm, each leg, front, back..."
    call sunscreenApplied 2
  else if deviceOn then
    call standbyUntilButton
  else
    call alreadyProtected
```

```

function
  set state to 0
  call tap
  if (de) then
    call show
  if (a) then
    call
  call st

```

extreme
UV 11 d
pressed
instead
sleeps

UV 8 to 10: flashes and beeps quickly until A is pressed, then shows the sunscreen advice plus hat and shade, and starts the 2 hour timer. It alerts again if the wearer so far only had the moderate advice.

ed: sunscreen is

Start the 2 hour reapply timer

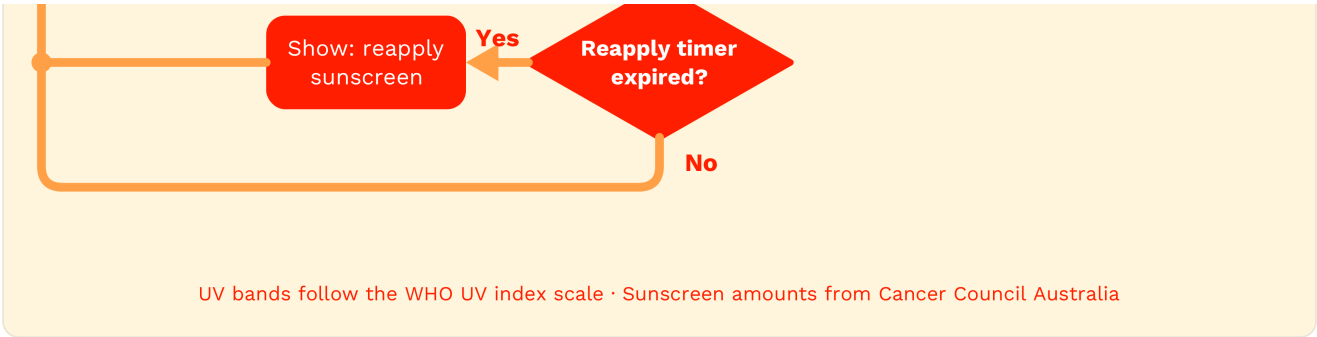
Sleep until any button is pressed

```
function start2HourReapplyTimer
  set reapplyTimerRunning to true
  set reapplyDueTime to running time (ms) + reapplyMs
```

Starts the 2 hour countdown to the next sunscreen.

and after a sunscreen
 numbers which advice was
 s one more application,
 , and starts the timer.

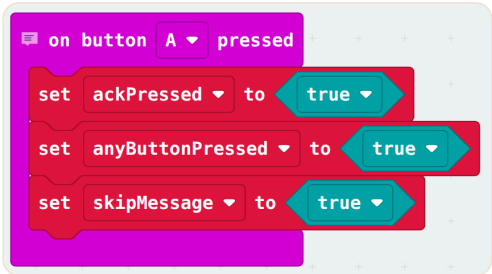
standbyUntilButton



The plan the device follows. Every box on it is a block on the left, with the same name.

The two buttons

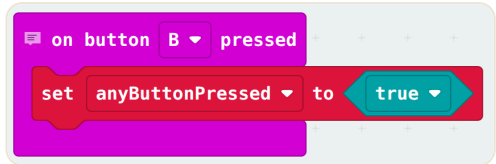
A: done



on button A pressed
Pressing A means done: sunscreen is on, or I am going inside. It also wakes the device, skips a long message, and during a wait shows the UV number and the countdown.

Pressing A and B together turns the device off, or on again (see the flowchart).

B: wake up



on button B pressed
Pressing B wakes the device from sleep. Holding B while switching on gives the fast demo timings.

The whole program



All 74 blocks in the editor, in nine sections. The flowchart on this sheet is sections 1 to 4; sections 5 to 9 are the helpers: screen, sound, Bluetooth, maths and the optional screen.

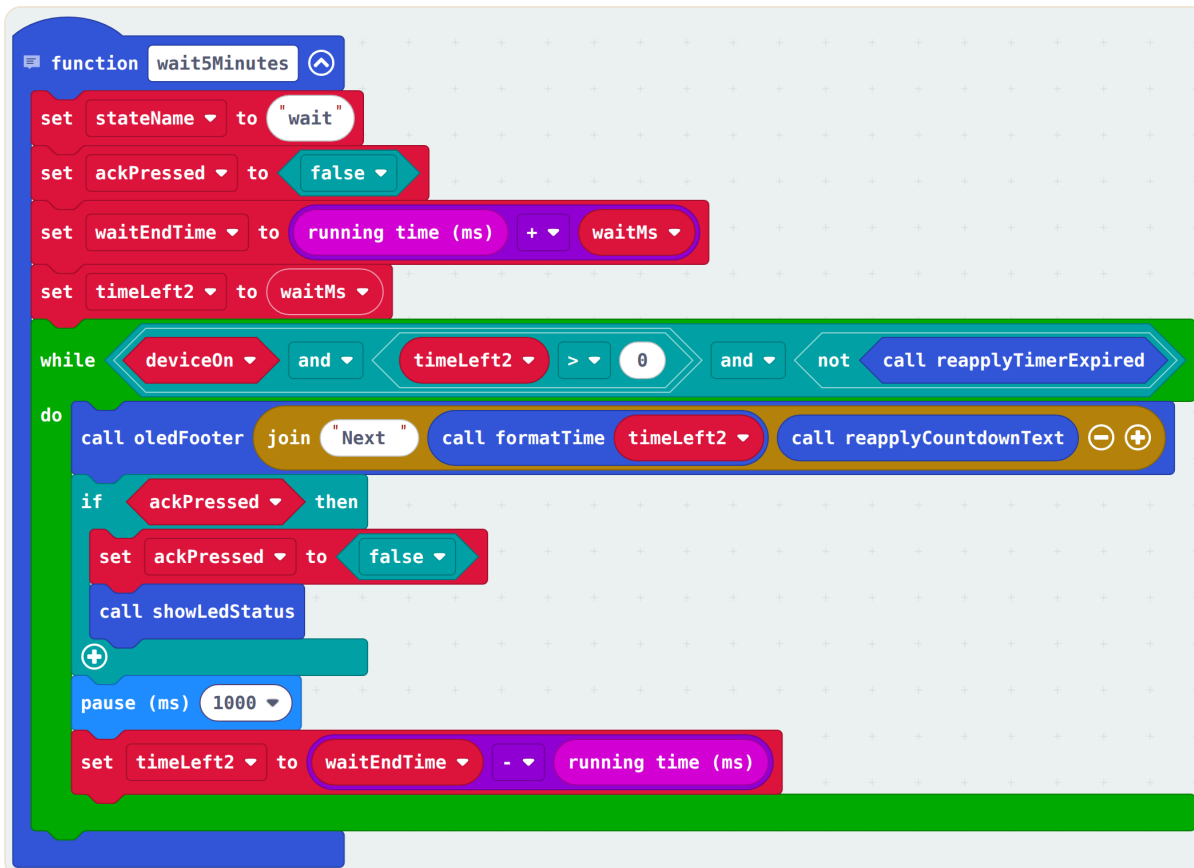
What the block colours mean

- | | |
|-------------------------------|-----------------------------|
| Variables: remember a value | Text |
| Functions: a step with a name | Arrays: lists |
| Basic: the lights, pauses | Music: beeps |
| Input: buttons and time | Pins: sensor, motor, screen |
| Loops: repeat | Bluetooth: the phone |
| Logic: if, compare | Serial: the USB cable |
| Maths | Control: timing |

Words used here

UV index How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or

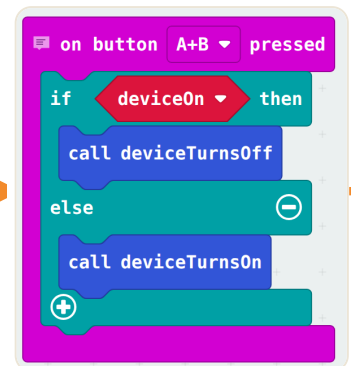
Wait 5 minutes



wait5Minutes

Waits 5 minutes before the next check. The wait ends early if the device is turned off or the sunscreen timer runs out. Pressing A during the wait shows the UV number and the countdown.

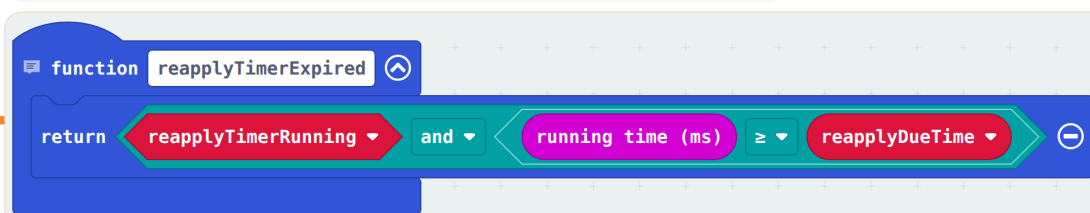
A and B held?



on button A+B press

Pressing A and B together device off, or on again.

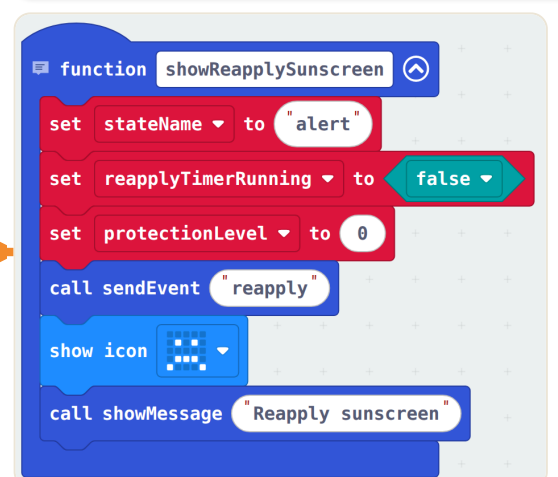
Is it time for more sunscreen?



reapplyTimerExpired

Asks: have the 2 hours since the last sunscreen passed?

Yes: show Reapply sunscreen



showReapplySunscreen

Shows Reapply sunscreen and clears the timer, so the next time round the flowchart alerts again the sun is still strong.

Sleep: waits for any button, or done from the app. Pressing A and B together still turns the device off. Afterwards it goes straight back to reading the sensor.

Yes: device turns off

```
function deviceTurnsOff
  set deviceOn to false
  set stateName to "off"
  call servoOff
  clear screen
  call oledClear
  call oledWriteRow 3 "OFF"
  call oledWriteRow 5 "A+B to turn on"
  call sendEvent "off"
```

deviceTurnsOff
Switching off: everything stops and the screen shows OFF. Bluetooth stays on so the app can switch it back on.

cloudy days.

micro:bit The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.

Block One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.

Sensor The part that measures UV. It gives a voltage, which the program turns into a UV index number.

Servo A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.

Bluetooth and USB The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.

OLED An optional small screen. Without one, the grid of lights does the job.

ms and Hz ms are milliseconds, thousandths of a second. Hz sets how high a beep sounds.

Demo timings Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.

Names in the blocks Words like uvSampleCount are the program's own names for its settings and memory. Read them as English: uv sample count is how many readings to take.

Every one of the 74 blocks, with its note, is in the booklet on the table.